Year 9 5.2 Yearly Exam Revision

Booklet B

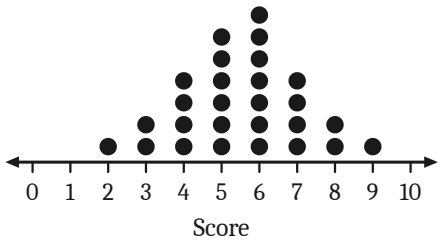
Answer every question in your **maths workbook**.
Show all working. A calculator is allowed.

Section 1: Data

1 Classify each type of data as **categorical** or **numerical**. If it is numerical, state whether it is **discrete** or **continuous**.

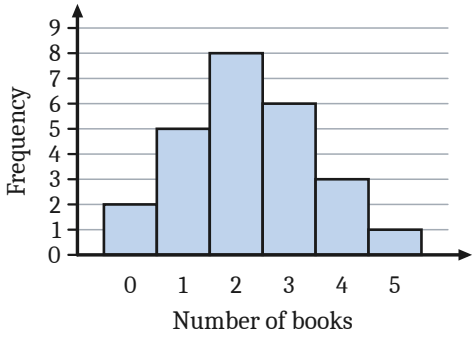
a the number of brothers and sisters
b the type of car a family owns
c the temperature at noon
d the number of goals scored in a game
e the length of a snake
f a rating of a movie (poor, good, excellent)

2 The dot plot shows the scores (out of 10) of students in a quiz.



a How many students are in the class?
b Describe the shape of the distribution (symmetrical, positively skewed or negatively skewed).
c Where do the scores cluster?
d Is there an outlier? If so, what is it?
e Find the mode.

3 The histogram shows the number of books each student in a group read last month.



a How many students are in the group?
b What is the mode?
c What is the range?
d How many students read fewer than 2 books?

4 Lucy recorded how many minutes late her bus was on 8 days:
4, 6, 3, 5, 2, 30, 4, 5

a Which score is an outlier?
b Find the mean and the median of all 8 scores.
c Find the mean and the median without the outlier.
d Which measure of centre is changed more by the outlier? Which one better describes the data?

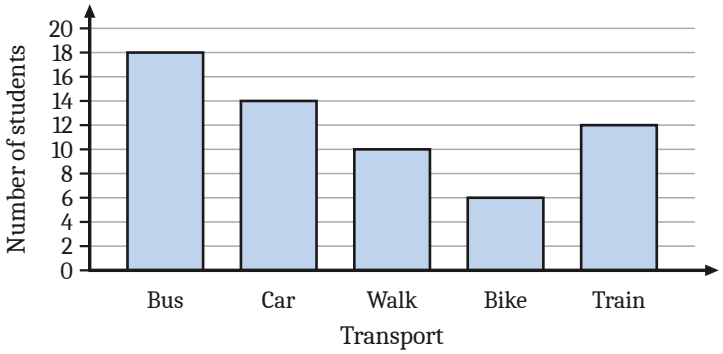
5 The stem-and-leaf plot shows how many push-ups 11 students did in one minute.

Stem	Leaf
0	8
1	2 5 7 9
2	1 1 4 8
3	0 3

Key: 2 | 4 = 24 push-ups

a Find the mean, correct to 1 decimal place.
b Find the median.
c Find the mode.
d Find the range.

6 Students were asked how they travel to school. The results are shown in the column graph.



a Which type of transport was the least popular?
b How many more students travel by bus than by car?
c How many students were surveyed?
d What percentage of the students walk or ride a bike? Answer correct to 1 decimal place.

7 The back-to-back stem-and-leaf plot shows the points scored by two basketball players in their last 11 games.

a Find the median for each player.

b Find the range for each player.

c Which player usually scores more points? Give a reason.

d Which player is more consistent? Explain.

Jai	Stem	Ben
	0	5 9
9 8 5 2	1	4 7
9 7 5 4 2 2 1	2	0 6 8
	3	1 3 6 8

Key: 2 | 4 = 24 points

Section 2: Surface area and volume

Formulas

1 kL = 1000 L, 1 L = 1000 mL, 1 L = 1000 cm³, 1 m³ = 1000 L. Absolute error = ½ × precision

Parallelogram $A = bh$; kite and rhombus $A = \frac{1}{2}xy$; trapezium $A = \frac{1}{2}h(a + b)$; triangle $A = \frac{1}{2}bh$

Circle $C = 2\pi r$, $A = \pi r^2$; cylinder $V = \pi r^2 h$, $SA = 2\pi r^2 + 2\pi rh$; rectangular prism $V = lwh$

8 Convert:

a 5.3 kL to L

b 1250 mL to L

c 3.75 L to mL

d 640 L to kL

e 7500 cm³ to L

f 0.8 m³ to L

9 For each measuring instrument, state (i) the precision and (ii) the absolute error.

a a tape measure marked in centimetres

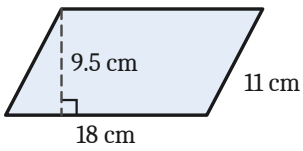
b a measuring cylinder marked every 50 mL

c bathroom scales marked every 0.5 kg

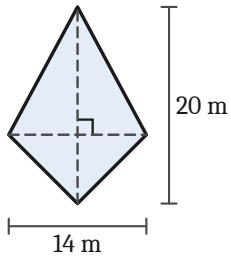
d a speedometer marked every 10 km/h

10 Find the area of each shape.

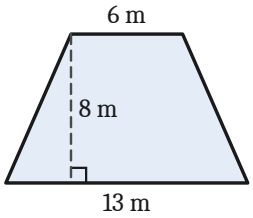
a



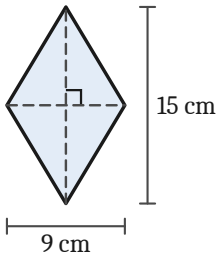
b



c

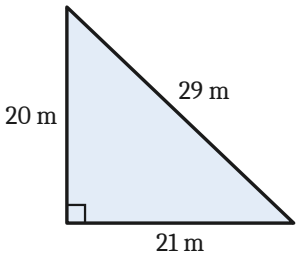


d

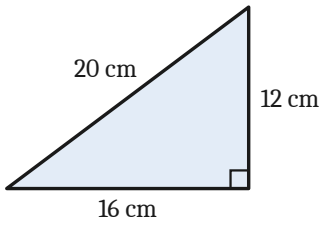


11 For each right-angled triangle, find (i) the perimeter and (ii) the area.

a

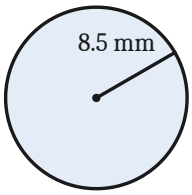


b

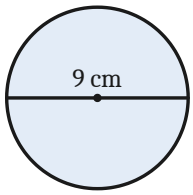


12 For each circle, find (i) the circumference and (ii) the area, correct to 1 decimal place.

a



b

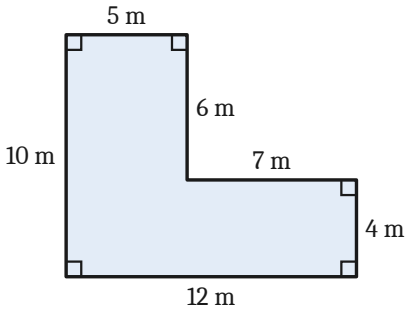


13 The plan shows the floor of a room. All of the corners are right angles.

a Skirting board is fitted around the edge of the room. How many metres are needed?

b Find the area of the floor.

c Carpet costs $45.90 per square metre. Find the cost of carpeting the whole floor.

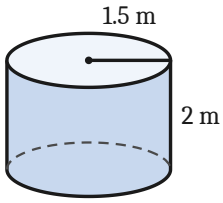


14 A closed rainwater tank is a cylinder with a radius of 1.5 m and a height of 2 m.

a Find the volume of the tank, correct to 2 decimal places.

b How many litres does the tank hold? Answer to the nearest litre.

c The outside of the tank is to be painted. Find its surface area, correct to 1 decimal place.

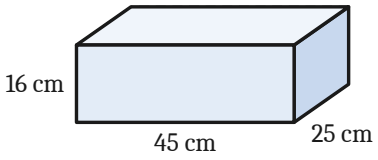


15 A closed toy box is a rectangular prism 45 cm long, 25 cm wide and 16 cm high.

a Find the volume of the box.

b How many litres does the box hold?

c Find the surface area of the box.



Section 3: Mixed topics

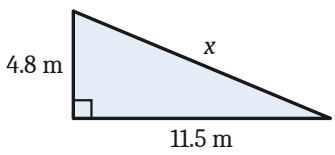
Formulas
Simple interest $I = Prn$; percentage profit = profit ÷ cost price × 100%; 1 year = 52 weeks = 26 fortnights = 12 months
Pythagoras' theorem $c^2 = a^2 + b^2$; $\sin\theta$ = opp ÷ hyp, $\cos\theta$ = adj ÷ hyp, $\tan\theta$ = opp ÷ adj

16 If $m = 5$ and $n = 2$, evaluate:

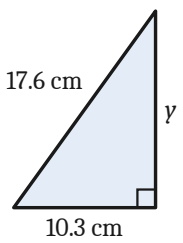
a $4m - n$ **b** $m^2 + 3n$ **c** $2n^2$ **d** $(m + n)^2$

17 Find the value of each pronumeral, correct to 1 decimal place.

a



b



18 Solve each problem.

a A pair of shoes marked at $120 is discounted by 25%. Find the sale price.

b A $360 phone is discounted by 12%. How much is saved, and what is the sale price?

c A shop buys a guitar for $280 and sells it for $378. Find the profit, and the profit as a percentage of the cost price.

19 Simplify, writing each answer in index notation.

a $4k^5 \times 7k^2$ **b** $18w^8 \div 3w^2$ **c** $(p^3)^4$ **d** $5m^2n^3 \times 3m^4n$

20 Noah is a teacher. He earns an annual salary of $92 040. How much is he paid:

a per week? **b** per fortnight? **c** per month?

21 Simplify:

a $8p - 3q + 2p + 7q$ **b** $5a \times 3c$ **c** $24xy \div 8y$ **d** $4t + 9 - t - 3$

22 Expand and simplify:

a $6(2m + 5)$ **b** $-3(4k - 7)$

c $3(a + 4) + 2(a - 5)$ **d** $4(3p - 2) - 2(p + 3)$

23 Factorise:

a $6k + 21$ **b** $14n - 35$ **c** $p^2 - 9p$ **d** $10x^2 + 15x$

24 Find the simple interest earned on each investment.

a \$8600 at 4.5% p.a. for 3 years **b** \$15 000 at 3.8% p.a. for 18 months

25 Solve each equation.

a $7x - 6 = 43$ **b** $x \div 4 + 3 = 8$ **c** $9a + 4 = 5a + 36$

d $2(3p + 4) = 4p + 20$ **e** $8n - 2(n + 5) = 3n + 11$

26 Sam sells furniture. He is paid \$480 per week plus 3.5% commission on his sales.

a Last week Sam sold \$36 200 of furniture. Find his total pay for the week.

b Mel earns 5% commission only (no retainer). Find her pay for a week in which she sells \$18 900 of goods.

27 Ava works at a supermarket. She is paid \$24.60 per hour for the first 38 hours each week and time-and-a-half for each hour after that. Last week she worked 41 hours.

a Find the pay for the first 38 hours.

b Find the pay for the overtime hours.

c Find the total pay for the week.

28 A taxi charges \$4.50 plus \$2.30 per kilometre. The cost, C dollars, for d km is $C = 4.5 + 2.3d$.

a Find the cost of a 12 km trip.

b Another trip cost \$50.50. Write an equation and solve it to find how far the taxi travelled.

29 A kite is flying on a string 25 m long. The kite is 18 m above the ground, directly above a point on level ground.

a Find the distance d along the ground, correct to 2 decimal places.

b Find the angle θ the string makes with the ground, correct to the nearest degree.

